

Saccade-Vergence Interactions in Humans

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SUMMARY AND CONCLUSIONS

1. We recorded eye movements in four normal human subjects during refixations between targets calling for various combinations of saccades and vergence. We confirmed and extended prior observations of 1) transient changes in horizontal ocular alignment during both pure horizontal saccades (relative divergence followed by relative convergence) and pure vertical saccades (usually divergence for upward and convergence for downward saccades); 2) occasional, high-frequency (20–25 Hz), conjugate oscillations along the axis orthogonal to the main saccade; and 3) the speeding up of horizontal vergence by both horizontal and vertical saccades.

2. To interpret these findings, we developed a hypothesis for the generation of vergence to step changes in target depth, both with and without associated saccades. The essential features of this hypothesis are 1) the transient changes in horizontal ocular alignment during pure horizontal saccades reflect asymmetries in the mechanical properties of the lateral and medial rectus muscles causing adduction to lag abduction; 2) pure vergence movements in response to step changes in target depth are generated by a neural network that uses a desired change in vergence position as its input command and instantaneous vergence motor error (the difference between the desired change and the actual change in vergence) to drive vergence premotor neurons; and 3) the facilitation of horizontal vergence by saccades arises from nonlinear interactions in central premotor circuits.

3. The hypothetical network for generating pure vergence to step changes in target depth is analogous in structure to the local feedback model for the generation of saccades and has the same conceptual appeal. With the assumption of a single nonlinearity describing the relationship between a vergence motor error signal and the output of the neurons that generate premotor vergence velocity commands, this model generates pure vergence movements with peak velocity–amplitude relationships and trajectories that closely match those of experimental data.

4. Several types of models are proposed for the central, nonlinear interaction that occurs when saccades and vergence are combined. Common to all models is the idea that omnidirectional pause neurons (OPN), which are thought to gate activity for saccade burst neurons, also gate activity for saccade-related vergence. In one model we hypothesize the existence of a separate class of saccade-related vergence burst neurons, which generate premotor horizontal vergence commands but only during saccades. In a second model we hypothesize separate right eye and left eye saccadic burst neurons that receive not only conjugate, but also equal but oppositely directed vergence error signals. In this way the difference between the outputs of the right eye and left eye saccade burst neurons produces a saccade-related horizontal vergence command. In a third model we propose that facilitation of vergence during saccades is a result of a multiplication (an increase in the gain of premotor vergence velocity neurons selectively during saccades).

5. The results of simulations of these models and comparison with our experimental data favor the first and third models, which either incorporate a separate class of saccade-related vergence burst neurons, or assume a change in property of premotor ver-

gence velocity neurons caused by the lifting of OPN inhibition during saccades. Simulations of these models lead to a number of predictions about the properties of neurons within the brain stem that generate vergence commands, both with and without associated saccades. Electrophysiological experiments are needed to confirm or refute these hypotheses.

INTRODUCTION

Saccadic and vergence eye movements are commonly treated as distinct subclasses of eye movements, with largely separate anatomic and physiological substrates, and control systems characteristics. In natural circumstances, however, when an abrupt change in the depth of the line of sight is required, vergence movements almost never occur without an associated saccade. Only in the laboratory, after considerable effort, can target stimuli be positioned precisely enough to provide a disparity signal alone, without any conjugate, “cyclopean” component, to elicit pure vergence movements. Even when the targets are correctly aligned to elicit pure vergence movements, saccades still occur frequently (Erkelens et al. 1989b; Levi et al. 1987).

In foveate animals, saccades and vergence are also linked by their common functional imperative: to bring images of objects of interest onto the fovea. Because pure vergence movements are much slower than saccades, if the two were simply combined, there would be a delay in bringing images to both foveae. Accordingly, it would be useful if the speed of vergence could be increased when vergence was combined with saccades. If this were the case, one might also predict some sharing of central circuitry to facilitate such an interaction.

Ono et al. (1978) and Kenyon et al. (1980b) pointed out that such an interaction between saccades and vergence does in fact occur; the movements of each eye are more unequal than one would predict from simple addition of a saccade command (of the same sign to each eye) and a vergence command (of opposite signs to each eye). As expected from this observation, the change in alignment when vergence is associated with a saccade is faster than the change in alignment when vergence occurs alone (Enright 1984, 1986; Erkelens et al. 1989b).

These observations raise the more general question of how the CNS generates saccades that are of different sizes in each eye; either during normal behavior when saccades and vergence are combined, or in special circumstances when the movements of each eye are of different sizes as a result of disconjugate saccade adaptation (Erkelens et al. 1989a; Lemij and Collewyn 1991a,b; Oohira and Zee 1991; Schor et al. 1990; Zee and Levi 1989). Is the mechanism for generating saccades of different sizes a specific property of the saccadic system, i.e., does the brain program saccades of

different sizes for the left and right eyes? Or is there an interaction between saccades and vergence to produce saccades that only appear to be programmed to be of different sizes for each eye?

To learn more about saccade-vergence interactions in normal circumstances, we recorded eye movements in four normal human subjects, during refixations between targets calling for various combinations of saccades and vergence. We confirmed prior reports of transient changes in horizontal alignment during pure horizontal and vertical saccades (Collewijn et al. 1988a,b; Kapoula et al. 1987), as well as a facilitation of horizontal vergence by both horizontal and vertical saccades (Enright 1984). In an attempt to understand these findings, we developed hypotheses for the generation of vergence to step changes in target position, either when vergence is made alone, or when vergence is combined with saccades. We will present simulations of several models of this saccade-vergence interaction and make predictions about the behavior of premotor neurons within the brain stem that generate saccade and vergence commands.

METHODS

Target stimuli

At eye level, along the midsagittal plane, a Plexiglas strip was placed in front of the subject. Three light-emitting diodes (LEDs), each about 1 mm diam, were positioned on the Plexiglas strip so as to call for changes in vergence of $\sim 2.5^\circ$, 5° , and 10° , relative to a 0° LED that was rear-projected onto a translucent tangent screen located at 1 m from the subject. Calculations were based on an interpupillary distance of 58 mm. Accordingly, the vergence angle when viewing the 0° LED on the tangent screen was $\sim 3.3^\circ$. LEDs were also projected onto the tangent screen so that the difference between them and the 0° LED subtended 2.5° , 5° , and 10° of arc, both horizontally and vertically. Because the targets were projected onto a flat surface, a small change in the vergence angle was also required for saccades made between the 0° LED and any of the other LEDs that were projected onto the tangent screen (e.g., 0.10° for a 10° horizontal saccade and 0.06° for a 10° vertical saccade). For practical purposes, we will refer to pairs of LEDs on the tangent screen as isovergence stimuli.

The alignment of the vergence targets in the midsagittal plane was first checked with the use of a subjective method. The position of the vergence array was adjusted until the subject, while fixing on the 0° LED on the tangent screen, reported that the positions of the two images of the LED on the vergence array overlapped the image of each of two LEDs displayed on the tangent screen at the correct, corresponding horizontal locations. For example, to align the target for a 5° vergence movement correctly, the subject fixed on the 0° LED on the tangent screen but paid attention to the location of the two images of the LED (for 5° vergence) on the vergence array relative to the image of each of the two LEDs located on the tangent screen at right and left 2.5° . The correct alignment of the targets was further confirmed by the lack of any net change in the conjugate position of the eyes during the recordings of pure vergence responses. The head of the subject was immobilized with a chin and forehead rest. All experiments were performed with the background dimly illuminated so that the subject could clearly see the LEDs but was also aware of the contours of the frame of the tangent screen and of the field coil system.

Eye movement recordings

Eye movements were measured with the use of binocular scleral annuli (Collewijn et al. 1975) with the magnetic field technique

(CNC Engineering). The horizontal and vertical positions of the right and the left eyes were recorded and, after being filtered through a six-pole, low-pass Bessel filter (180 Hz), sampled at 500 Hz, with 12-bit precision, by a digital computer. The measurements were calibrated with the use of 10° horizontal and vertical target displacements. Effort was made to eliminate cross talk between the horizontal and vertical channels by phase adjustments following saccades between pure horizontal and pure vertical stimuli. All recordings were made with subjects viewing out of both eyes while wearing their corrective spectacles.

Experimental paradigms

Trials were collected in blocks of 12 back and forth movements between a pair of LEDs. Saccades and vergence were always triggered by the disappearance of one LED and the appearance of the other. The timing of target appearance was randomized (with a range of 1,750–2,750 ms), but the locations of the two LEDs were predictable after the first trial in each set. Pure saccades were elicited by illuminating the 0° LED on the tangent screen, and then one of the other LEDs on the tangent screen. Then the 0° LED was reilluminated, and the sequence repeated for 12 cycles. Each trial in which a vergence movement was required started at one particular LED on the vergence array. The next stimulus always appeared on the tangent screen, either at 0° , to elicit a pure divergence movement, or at one of the other LEDs projected onto the tangent screen, to elicit a saccade combined with divergence. The same LED on the vergence array was then reilluminated to elicit either a pure convergence movement, if the LED on the tangent screen had been at 0° , or a saccade combined with convergence, if one of the other LEDs on the tangent screen had been illuminated. In this way, a series of 2.5° , 5° , and 10° vergence movements was acquired, both alone, and combined with vertical and horizontal saccades of 2.5° , 5° , and 10° .

All trials calling for a specific amplitude of change in vergence were combined together in a single block. Then, the position of the near target was changed and another block of trials collected. Subjects were instructed to move their eyes as quickly as possible whenever the LED appeared, but not to anticipate the next target position. Subjects were also instructed not to blink during the change in fixation. The entire data set was obtained in two 35-min sessions on separate days for each subject.

Data analysis

Data analysis was performed off-line with an interactive program in which each individual trial was displayed on a video monitor. The position of each eye was shown as a function of time during the trial (for 1,200–1,500 ms after the displacement of the target). To show the change in the vergence angle, the horizontal position of the left eye was subtracted from the right eye. Accordingly, divergence was always positive, convergence negative. To show any change in the horizontal, conjugate position of the cyclopean eye, the horizontal positions of the right and left eyes were averaged. For our purposes, the term conjugate position (or velocity) will always refer to the average of the horizontal positions (or velocities) of the left and right eyes. The individual eye movement traces, as well as the derived vergence and conjugate traces, were also differentiated with a finite impulse response filter of length 15 (a duration of 30 ms) with bandwidth (-3 db) of 21.5 Hz to obtain the velocity of each eye as well as of the vergence and conjugate positions of the eyes. The filter was of the repeated-central difference type with $N = 4$ and $L = 3$ (Usui and Amidror 1982). Traces were ignored in which fixation was not steady at the beginning of the trial, in which blinks appeared to have occurred during the vergence response, or in which the initial saccade or conjugate change was less than one-half of the required amplitude.

Each individual trial was marked by identifying the beginning of the saccade (designated as "i," the initial position), either on the vertical eye position trace, for vertical saccade trials (with or without vergence), or on the horizontal conjugate trace, for horizontal saccade trials (with or without vergence; see, for example, Fig. 1A). The end of the rapid pulse portion of the saccade, or of the conjugate trace was also identified (designated as "p," the final position). These points were initially guessed by a computer algorithm with the use of velocity and acceleration criteria and a template for the saccade trajectory (Waitzman et al. 1991), but the placement of the marks was always verified by the experimenter for accuracy. If there was any rapid reversal (i.e., a "dynamic overshoot") at the end of the saccade, the p was placed at the beginning of that reversal. Otherwise, the end of the pulse portion of the saccade usually corresponded to conjugate (or vertical) eye velocity decreasing to $<40^\circ/\text{s}$.

For trials of pure saccades and saccades combined with vergence, the vergence traces were marked at the same two points in time as the i and p on the conjugate (or vertical) trace so that a measure of any change in alignment during the saccade (p minus i) could be obtained (see Fig. 1A). The duration of vertical saccades or of the conjugate traces for horizontal saccades was taken as the difference between the times of placement of the p and the i. The peak values of both conjugate (or vertical) and horizontal vergence velocity were also recorded. For trials in which vergence was called for (with or without saccades), the beginning and the end of the vergence movements were identified by an algorithm based on a vergence template, in a similar way to that for saccades.

Many "pure" vergence trials were "contaminated" by saccades [defined here as $>0.25^\circ$ of a rapid change (completed in <100 ms) in the conjugate (or vertical) position of the eyes during the vergence movement]. These seemingly extraneous saccades occurred even though there was little conjugate retinal error (i.e., no target misalignment) as verified by the observation that the conjugate position of the eyes was the same at the beginning and at the end of the trial. In most cases, pure vergence trials that were contaminated with saccades were excluded from further quantitative analysis unless the saccade occurred near the end of the vergence move-

ment, in which case it would not interfere with the analysis of the vergence movement near the time of vergence peak velocity.

There was often a rapid change in the horizontal conjugate position of the eyes during pure vertical saccades or when combined with horizontal vergence movements (see also RESULTS and Fig. 5). In many instances, this horizontal motion of the conjugate position appeared to be part of an oscillation because, by the end of the vertical saccade, the horizontal conjugate trace had usually returned near to the same position that it was at the beginning of the saccade. If, however, at the end of the horizontal conjugate oscillation there was a *net* change in horizontal conjugate position that was $>0.25^\circ$, the trial was excluded from the analysis. If the change was $<0.25^\circ$, it was retained in the analysis.

To compare the amplitude of the change in intrasaccadic alignment that occurred when vergence was combined with saccades, with the change when vergence occurred alone, we analyzed our data in the following way. First, we quantified the net change in alignment during pure saccades and the net change in alignment during saccades combined with vergence. We measured the change in ocular alignment during the saccade (defined by p minus i on the conjugate trace), by placing the same marks on the vergence trace and measuring their difference (see Fig. 1A). The amount of the change in alignment during pure saccades was then subtracted from the amount during the vergence combined with a saccade, to obtain an estimate of the component of the change in ocular alignment during the combined response that might be attributed to the vergence system.

We next calculated the change in alignment that occurred during a pure vergence movement in a time period corresponding to the duration of the conjugate change (p minus i) during the combined vergence-saccade response (Fig. 1B). The time period for this calculation was centered around the point of maximum vergence velocity during the pure vergence movement. The value of the change in alignment during this epoch of pure vergence became our estimate of what the contribution from the pure vergence system would have been during the response in which vergence was combined with a saccade. For subjects in whom pure vergence responses were contaminated by saccades (primarily subject 4, for 5 and 10° divergence) we had to estimate the contribution from the pure vergence system by extrapolating from portions of the pure vergence trace that were not affected by the saccade.

Subjects

Four healthy subjects (S1–S4), 21–47 yr of age, were tested. Their corrected visual acuities were 20/20 or better, with three subjects requiring and wearing a spectacle correction: S1, OD +2.25, OS +2.25; S2, OD $-7 +1.25 \times 10$, OS -7.50 sphere; S3, OD $-4.25 +1.75 \times 11$, OS $-3.50 +1.50 \times 165$. By a simple sighting test, through one's closed thumb and index finger, subjects S1 and S3 appeared right eye dominant and S2 and S4 left eye dominant.

RESULTS

Here we characterize the general features of the eye movement responses to the various combinations of saccade and vergence target stimuli. More subtle differences in the eye movement trajectories, which are particularly germane to the model simulations, will be considered in the DISCUSSION.

Pure vergence responses

Figure 2 shows profiles of vergence position and vergence velocity for convergence and divergence responses to 5°

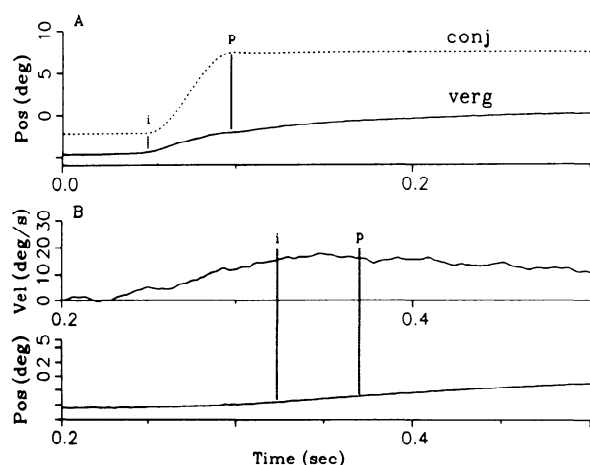


FIG. 1. Vergence combined with saccade (A) and pure vergence (B) to illustrate the method of data analysis. An "i" and "p" were placed on the conjugate position trace (---) to identify the beginning and the end of the saccade, and then projected onto the vergence trace (—) so that the intrasaccadic change in alignment could be quantified. For comparing the alignment change during vergence combined with saccades, and the change that would have occurred in the same time period during pure vergence movements, an i and p were placed on the pure vergence trace using the time-of-peak vergence velocity as the middle of an epoch, with the same duration as the conjugate change during the combined response (B, top and bottom traces). See also text.

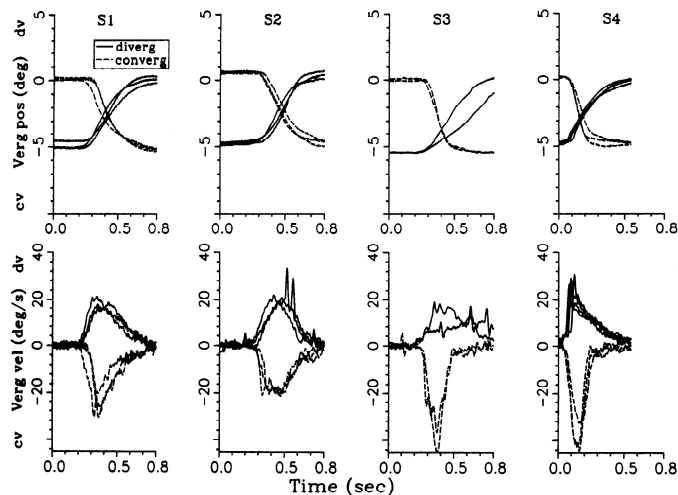


FIG. 2. Divergence and convergence responses for all 4 subjects (*S1–S4*) to 5° pure disparity stimuli. Two or 3 responses in each direction are superimposed for each subject. Vergence position is depicted in the *top* traces, vergence velocity in the *bottom* traces. Divergence is positive, convergence is negative. Note that the peak velocity of pure convergence was usually faster than that of pure divergence. Upward spikes in the divergence velocity traces reflect a boost in vergence speed from coincident saccades (e.g., *S2*). In this and subsequent figures the data were often offset horizontally for clarity so 0 on the time scale does not necessarily correspond to the beginning of the trial when the data were collected.

disparity stimuli for all subjects. In all cases two or three trials are superimposed. In some traces there were small spikelike increases in vergence velocity that reflect the effects of coincident saccades (e.g., the records of *subjects 2* and *4*). For the entire range of vergence amplitudes elicited (2.5–10°), the mean values for peak velocity of pure convergence movements were always faster than for pure divergence with the exception of *subject 2* for the 5° vergence trials, in which case they were about the same (Fig. 2, *2nd panel*). The lowest values of mean peak velocity during pure vergence, for a given subject, ranged between 9.5 and 13°/s for 2.5° divergence and the highest, between 41 and 58°/s for 10° convergence. In the case of *subject 4*, a value for peak vergence velocity for the 5 and 10° divergence trials was extrapolated from the portions of the vergence response that were not contaminated by saccades. For each of our subjects, the vergence waveforms were usually remarkably stereotyped from trial to trial, although some variability was apparent [e.g., *subject 3* (*S3*) for divergence].

Figure 3 shows the vergence velocity profile for a group of divergence and convergence responses of 2.5, 5, and 10° in amplitude from *subject 1*. Note that not only is the peak velocity of pure divergence less than for convergence but that the overall vergence response appears more sluggish during divergence than during convergence.

Many of the responses between targets that were positioned to elicit a pure vergence response also contained saccades (Fig. 4). The saccade could not be attributed to the presence of a cyclopean retinal error, because the saccade was usually larger than one would expect from any possible small degree of target misalignment. Note that the conjugate positions of the eyes at the beginning and at the end of trial are nearly identical (Fig. 4, conjugate trace). In all subjects, the seemingly inappropriate saccades occurred

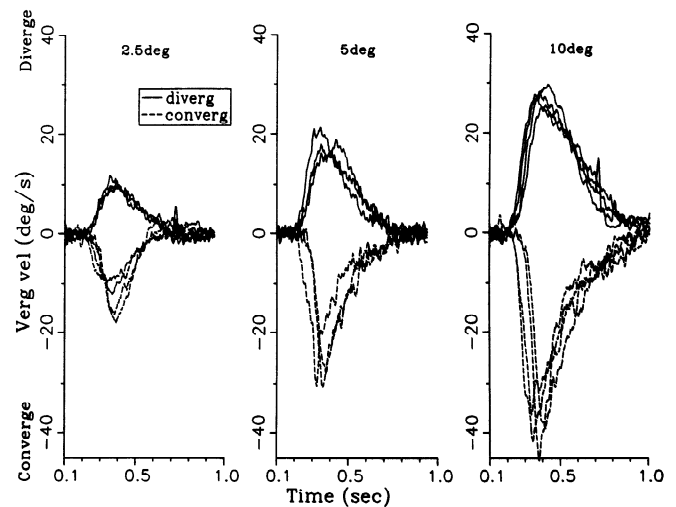


FIG. 3. Vergence velocity traces for several superimposed responses of 1 subject (*S1*) to pure disparity stimuli of 2.5, 5, and 10°. Divergence is positive, convergence is negative. The occasional sharp spikes in the velocity traces (e.g., toward the end of the 2.5° divergence trace) reflect small coincident saccades (see text). In this subject, peak velocity, and the rates of rise to and fall from peak velocity, were higher for convergence than for divergence.

more commonly during divergence, and especially for the larger amplitudes. The direction, amplitude, and frequency of occurrence of saccades during pure vergence, however, were idiosyncratic among the four subjects. With pure vergence stimuli, saccades were most frequent and of the largest amplitude in *S4*, shown in Fig. 4.

When saccades did occur with pure vergence stimuli, one eye would be taken much closer to the target at the expense of the other eye (Fig. 4, *top traces*). Then the eye that had

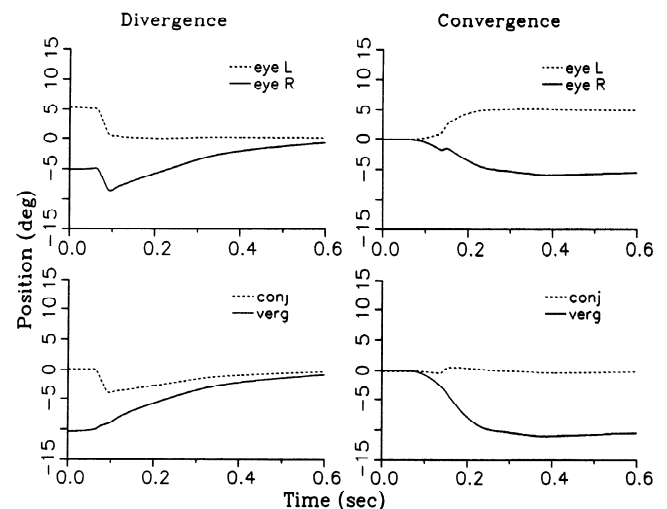


FIG. 4. Responses to 10° divergence and convergence disparity stimuli of *subject 4*. Divergence is positive, convergence is negative. This subject appears to be strongly left eye dominant, and, in both responses, but especially for divergence, the left eye is preferentially taken toward the target 1st. Note the asymmetric vergence response (*top traces*, compare left and right eyes) and the smooth drift of the conjugate position of the eyes (*bottom traces*) after the saccade. Conjugate position of the eyes is nearly identical at beginning and end of trial, indicating that target misalignment could not have accounted for presence of the saccade.

been taken away from the target would be brought to the target slowly with an asymmetric or even monocular vergence movement (Fig. 4, *top traces*). At the same time there was a smooth drift of the conjugate position of the eyes, which corrected for the initial saccade that took the conjugate position away from the target (Fig. 4, *bottom traces*).

Pure saccade responses

During both pure horizontal and pure vertical saccades, there were transient changes in horizontal ocular alignment. For *horizontal saccades*, there was a relative divergence of the visual axes during the initial portion of the saccade and then a relative convergence of the axes until the eyes became correctly aligned, usually at ~ 30 – 100 ms after the end of the saccade (e.g., Fig. 6, *A and B*, 2nd panels). During *vertical saccades*, there was also a transient change in horizontal alignment; usually an initial divergence for upward saccades and an initial convergence for downward saccades (e.g., Fig. 7 *B*, *top panels*).

We also noted on some trials that there was a change in the conjugate position of the eyes along the axis orthogonal to the target displacement, i.e., a horizontal conjugate change during pure vertical saccades (Fig. 5 *A*), or a vertical conjugate change during pure horizontal saccades. Conjugate horizontal oscillations also occurred when vertical saccades were combined with horizontal vergence (Fig. 5 *B*). The conjugate change in alignment along the orthogonal axis was best appreciated on the conjugate velocity traces

(Fig. 5, *A and B*, *top right traces*). The conjugate motion of the eyes on the orthogonal axis could be interpreted as part of a cycle of an oscillation with a frequency of ~ 20 – 25 Hz.

Saccades combined with vergence

Typical responses in which saccades and vergence are combined are shown in the third panel (Combined) of Fig. 6, *A and B*, for *subject 1*. The *top traces* show vergence and/or conjugate position, and the *bottom traces* show vergence velocity for, respectively, a 5° pure vergence movement (*far left panels*), a 10° pure saccade (*2nd panels*), a combined 5° vergence with 10° saccade (*3rd panels*), and a comparison of the changes in alignment between the actual combined response and what would be the sum of a pure vergence and a pure saccade (*far right panels*). Simple linear addition of the alignment change that occurs during the pure saccade and the pure vergence movement does not seem to account for the changes during vergence combined with saccades (*far right panels*).

Figure 7, *A and B*, shows a number of combinations of 2.5° vergence with saccades of different sizes and directions for *subject 2*. The velocity of the alignment change with a pure saccade (*top traces*) is compared with the vergence velocity during vergence combined with saccades (*bottom traces*). The velocity of a pure vergence response (*far left*) is also shown for comparison. For both vertical and horizontal saccades, the waveform of vergence velocity during the combined vergence-saccade movement was not what one would predict from a linear combination of contributions

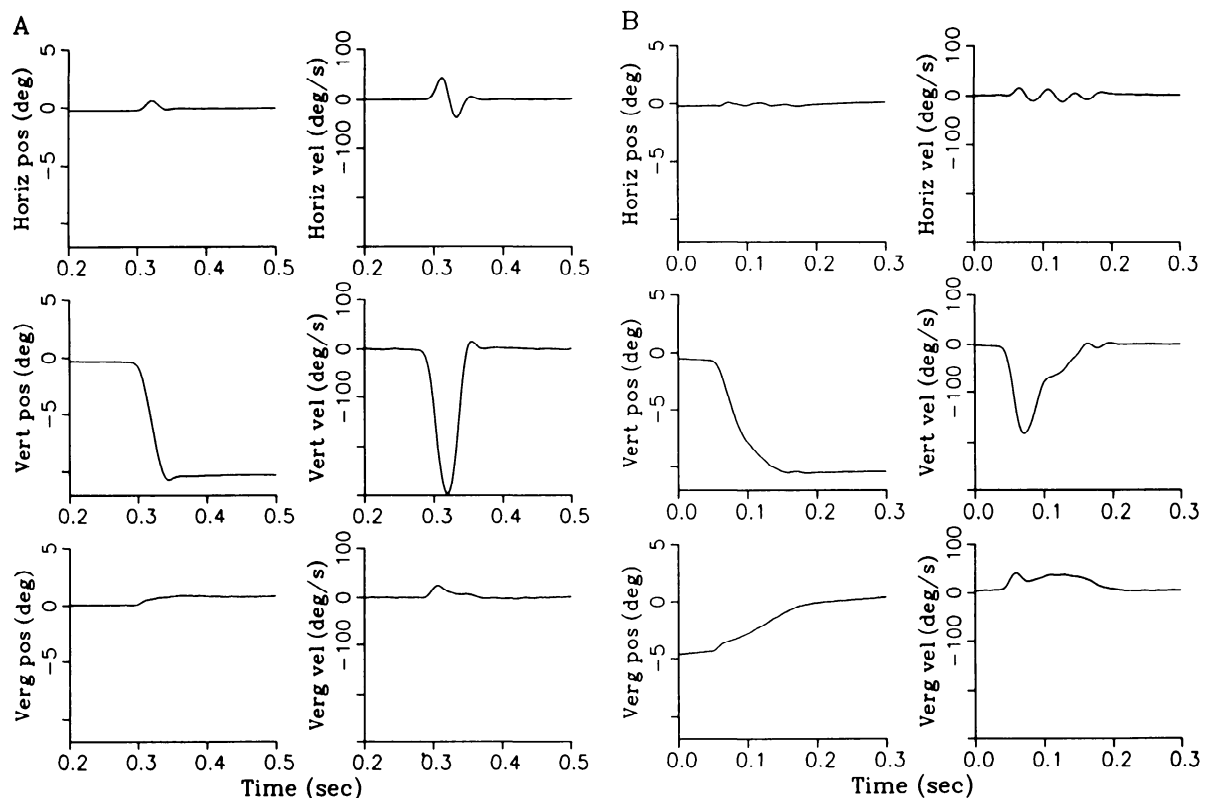


FIG. 5. Conjugate horizontal oscillations associated with a pure 10° vertical saccade (*A*) and with a 10° vertical saccade combined with a 5° vergence (*B*); (*subject 4*). Horizontal oscillations have a frequency of ~ 20 – 25 Hz. Note there is virtually no net change in horizontal conjugate position from the beginning to end of vertical saccade. The several cycles of horizontal oscillations with the combined vergence-saccade (*B*) are associated with a slower and longer-duration vertical saccade.

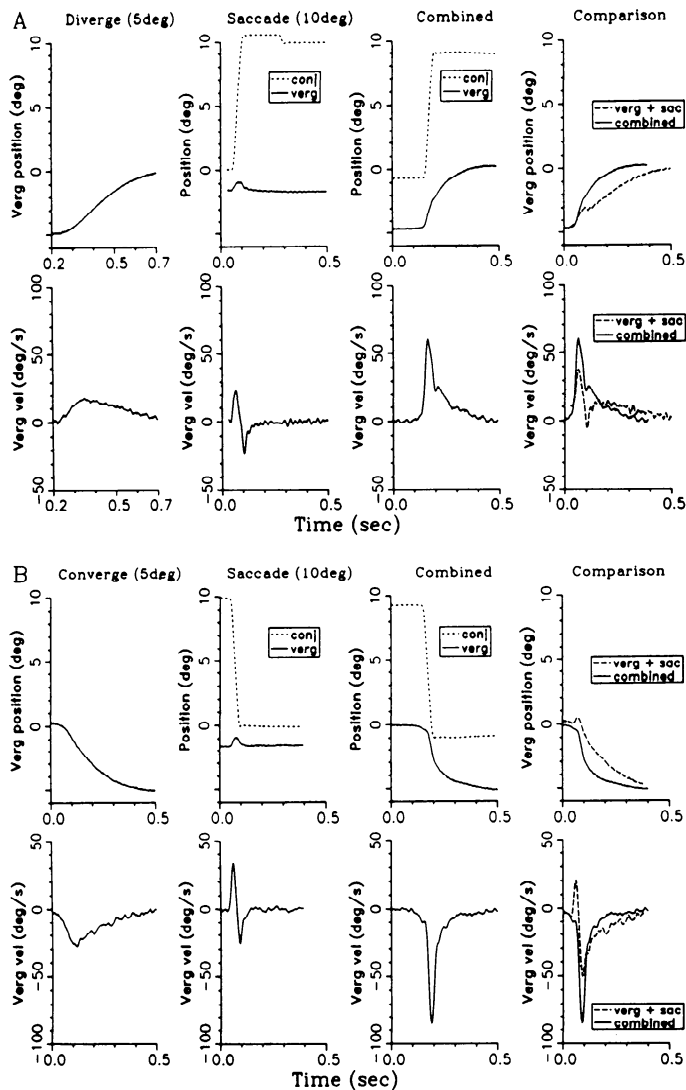


FIG. 6. Comparison of responses of *subject 1* to a stimulus calling for a 10° saccade combined with 5° of vergence, a 5° pure vergence, and a 10° pure saccade. *Top*: from left to right, pure vergence response, pure saccade response, combined response, and comparison traces between the pure vergence plus pure saccade response (---) and the combined response (—). *Bottom*: vergence velocity traces for the same responses shown in the *top* traces. *A*: divergence responses. *B*: convergence responses. Note that the combined vergence-saccade responses can not be accounted for by the sum of the alignment change from a pure saccade and from a pure vergence movement (comparison panel).

from the change in alignment associated with pure saccades and with pure vergence responses.

Typical responses in which saccade and vergence movements are combined are shown in Fig. 8 for all four subjects. The *top* traces depict vergence velocity and the *bottom* traces conjugate velocity. Note that the initial positions of the vergence velocity waveform reflect the inherent divergence associated with the beginning of horizontal saccades. The general shape of the vergence velocity traces was similar among all four subjects. There was more variability among subjects, however, for the vergence velocity than for the conjugate velocity waveforms (compare *top* and *bottom* traces).

To confirm that the amplitude of the change in intrasaccadic alignment that occurred when vergence was com-

bined with saccades was above what one would expect from simple addition of the component parts from pure saccades and pure vergence, we quantified the net change in alignment during pure saccades, the net change in alignment during saccades combined with vergence, and the expected change from pure vergence acting alone (see METHODS). The results of this analysis for all four subjects, for both horizontal and vertical 10° saccades combined with 2.5° vergence movements, are shown in Fig. 9, *top*. Absolute values for the estimate of the contribution from the pure vergence system (stippled bars) are compared with the estimate of the contribution from the vergence system during the combined response (solid bars). Thus a positive differ-

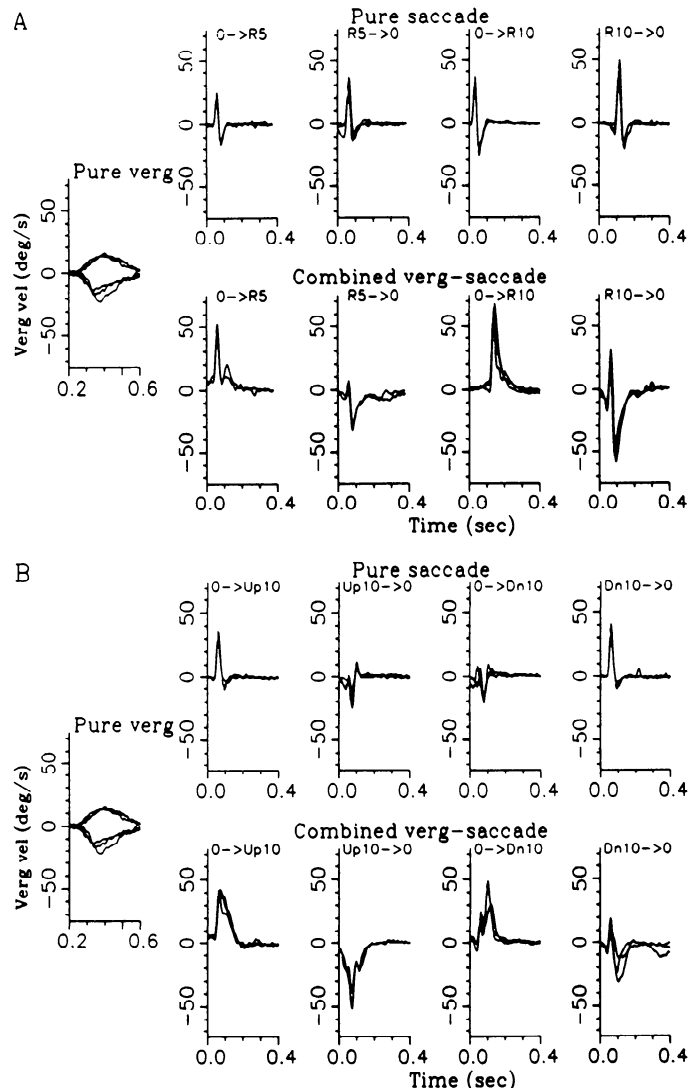


FIG. 7. Comparison of responses of *subject 2* to stimuli calling for 2.5° of vergence combined with saccades of varying sizes and direction. Saccades away from 0 are combined with divergence, and saccades toward 0 are combined with convergence. Divergence is positive, and convergence is negative. *Top*: vergence velocity during pure saccades. *Far left*: vergence velocity during a pure 2.5° vergence movement. *Bottom*: vergence velocity during 2.5° vergence combined with saccades of the size and direction indicated. Several trials of the same type are overlapped. Vergence velocity traces indicate that the combined response is not a simple sum of the alignment changes from the pure saccade and the pure vergence movement. *A*: horizontal vergence with horizontal saccades. *B*: horizontal vergence with vertical saccades.

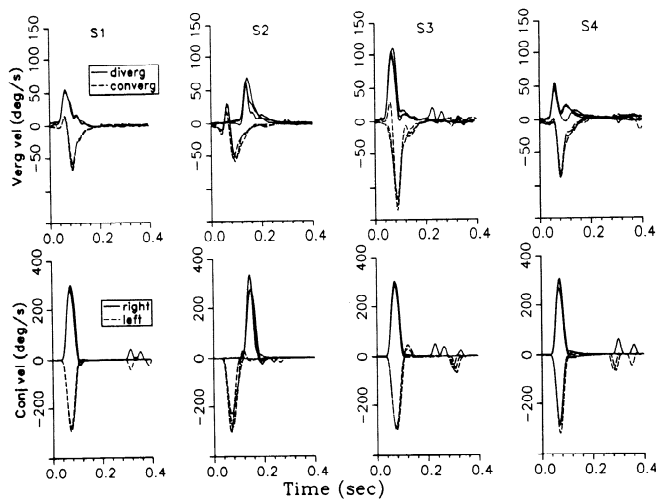


FIG. 8. Combined responses to stimuli calling for a 10° horizontal saccade with 2.5° vergence are shown for all subjects (S1–S4). *Top*: vergence velocity. *Bottom*: horizontal conjugate velocity. Divergence and right are positive, convergence and left are negative. Rightward saccades occur with divergence and leftward saccades with convergence. Transient divergence associated with horizontal saccades is reflected in the initial portions of the vergence velocity traces for both divergence and convergence. Note that the general shape of the waveforms of vergence velocity is similar among all subjects. The records of S2 have been offset for clarity.

ence between the solid bar and the stippled bar reflects the amount of facilitation of vergence by saccades. In most cases, the value of the contribution from the vergence system during the combined response was greater than the estimate of the contribution from the pure vergence system, although there was considerable variability among subjects. For all four subjects, facilitation was slightly greater with horizontal than with vertical saccades. Figure 9, *bottom*, shows a similar analysis for saccades of all sizes made with 2.5° vergence movements for one subject (S1). Again, facilitation of vergence by saccades is clearly evident. Similar results were obtained for saccades combined with 5 and 10° vergence movements (see Fig. 21, APPENDIX).

DISCUSSION

Experimental findings

Our results concur with previous studies of pure vergence and of vergence combined with saccades. Pure vergence responses were frequently punctuated by saccades, even when the target configuration did not call for any change in the conjugate position of the line of sight (Erkelens et al. 1989b; Kenyon et al. 1980a; Levi et al. 1987). There were transient changes in horizontal alignment during both horizontal and vertical saccades (Kapoula et al. 1987; Enright 1989; Collewijn et al. 1988a,b). Both horizontal and vertical saccades facilitated horizontal vergence, although more so with horizontal saccades (Enright 1984, 1986). A few features from our results, however, deserve further emphasis.

SACCADES WITH PURE VERGENCE. The saccades that occur during pure vergence can be interpreted as part of a strategy to take one eye, perhaps the dominant one, to the target promptly, but at the expense of the other eye being taken away from the target. A similar phenomenon occurs when

vergence is combined with saccades, as described by Ono and Tam (1981). After the saccade, the eye that is driven away from the target is slowly brought back to the target with what appears to be a markedly asymmetric or even a monocular vergence movement.

The reason underlying this pattern of behavior is not clear, and the degree to which it occurred was idiosyncratic from subject to subject. Such a strategy, however, would help to ensure the prompt identification, by at least one eye, of targets that suddenly appear away from the point of regard. The calculation of the precise location in depth and the exact configuration of the target would still have to await completion of vergence and binocular fixation. The observation that this asymmetric behavior occurred more frequently with divergence may relate to the fact that new targets that require immediate identification usually appear in the visual space beyond, rather than inside, the point of regard. Furthermore, stereopsis makes its most important contribution to depth perception when targets are relatively close. Other, monocular cues, such as relative size and parallax, are probably more important for localizing more distant objects.

Vergence responses are also known to be highly dependent on the nature of the task and the characteristics of the vergence stimuli (e.g., Erkelens 1987; Erkelens et al. 1989c). For example, larger stimuli tend to elicit faster vergence movements, and vergence during body movement toward stationary targets is faster than vergence made to moving targets with the body still. Yet, for a fixed set of stimuli calling for a step change in vergence position, with or without saccades, our subjects showed remarkably stereo-

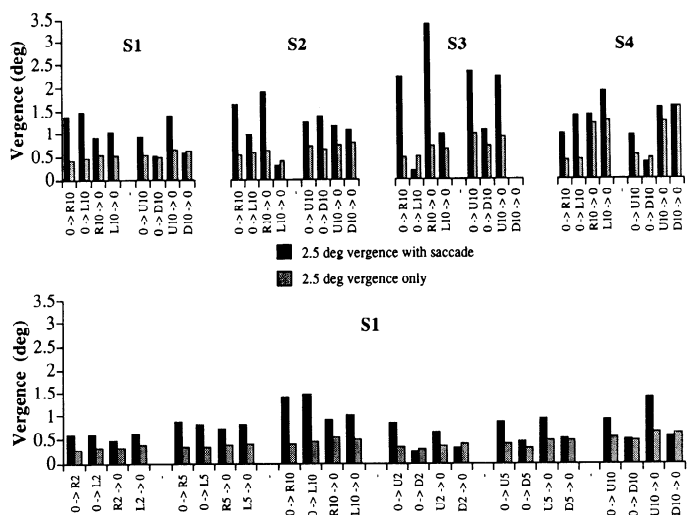
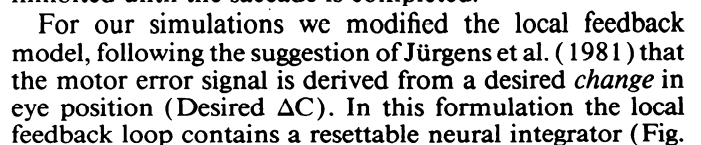


FIG. 9. Facilitation of vergence by saccades. Saccades away from 0 are combined with divergence, and saccades toward 0 are combined with convergence. Difference between the solid bar (combined response) and the stippled bar (pure vergence contributions) reflects the amount of facilitation. *Top*: response to stimuli calling for 2.5° of vergence combined with a 10° saccade is shown for all subjects (S1–S4). *Bottom*: response to stimuli calling for 2.5° of vergence with varying sizes of saccades is shown for subject 1. In most cases each histogram bar reflects the mean of at least 9 trials. Saccade sizes were 2.5 (abbreviated as 2), 5, and 10 degrees. Standard deviations of the values for the intrasaccadic vergence for subject 1 are shown in Fig. 21 (average standard deviation for all trial types was 19% of the mean) and were typical for all 4 subjects. See text for the specific description of the analysis procedure.

Local feedback model. Our template for a simulation of the generation of pure saccades is the local feedback model, originally proposed by Robinson and colleagues (Fig. 10A) (Robinson 1975; Zee et al. 1976). In this scheme a motor error signal (CME, the difference between desired and actual eye position) drives brain stem burst neurons to produce the necessary premotor saccade velocity commands (CVC). The saccade velocity command is passed directly to the motoneurons (the pulse of innervation), along with an integrated (the step of innervation) and a filtered (the slide of innervation) version, in correct proportions to compensate for the dynamic properties of the ocular motor plant (Optican and Miles 1985). In the local feedback model, the triggering of saccades is controlled by omnidirectional pause neurons (OPN); they discharge tonically during fixation and cease discharging (pause) during saccades



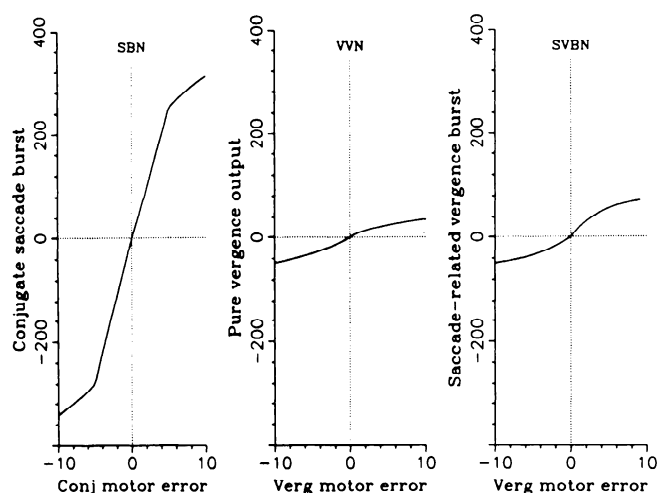


FIG. 11. Nonlinearities used in the simulation of pure saccades (*left*), pure vergence (*middle*), and vergence combined with saccades for the SVBN model (*right*). In each case the input is motor error [either conjugate (*left*) or vergence (*middle* and *right*)], and the output is the firing rate of the conjugate saccade burst neurons (SBN), vergence velocity neurons (VVN), or the saccade-related vergence burst neurons (SVBN). Note the slight asymmetries between the nonlinearities for divergence (positive) and convergence (negative). Equations and specific parameters describing these nonlinearities are in Table 1. Equations were chosen empirically to produce a nonlinearity that fits the experimental data.

10A, CRI) so that an instantaneous efference copy of the change in eye position ($\Delta C'$, derived from the integral of the premotor saccade velocity command) is fed back and subtracted from the desired change in eye position signal, to create an instantaneous motor error signal (CME). It is this signal that sustains the discharge of the saccade burst neurons. The exact saccade trajectory and the corresponding relationship between peak saccade velocity and saccade amplitude (the "main sequence") are determined by 1) the shape of the nonlinear relationship between motor error and the output of the burst neurons (see Fig. 11, SBN); 2) the matching of the pulse, slide, and step components of saccade innervation; and 3) the characteristics of the ocular motor plant.

We combined this local feedback model of the saccadic pulse generator with a fourth-order linear approximation to the ocular motor plant containing one zero, two real poles and two complex poles (Optican and Miles 1985; Robinson 1964) (see APPENDIX, Fig. 19, and Table 1).¹ This model produced realistically appearing saccades as reflected both in peak velocity-amplitude relationships and in the analysis of saccade trajectories by phase planes (plots of instantaneous values of eye position versus eye velocity; Fig. 12). The phase plane diagrams in particular reflect the nuances of the saccade trajectory and demonstrate how well the model matches the data (Fig. 12, *bottom*).

Disconjugacy during horizontal saccades. To simulate the transient disconjugacy of horizontal saccades made between isovergence targets, we propose that the mechanical

properties of the lateral and medial rectus muscles differ (Collewijn et al. 1988a). This assumption seems plausible because their sizes are different. On the basis of the observation that one eye seemed to lag the other, we empirically adjusted the time constant of the plant model pole that corresponds to the faster of the serial viscoelastic elements (see also APPENDIX). With the choice of a difference of 3.0 ms in time constant, depending on whether the eye was abducting (time constant of 10 ms) or adducting (13 ms), the simulation closely matched the experimental data (Fig. 13, *left*).

Another way to simulate the change in alignment during pure horizontal saccades is to delay (by 1.9 ms) the arrival of premotor signals at medial rectus motoneurons relative to the time of their arrival at lateral rectus motoneurons. This difference in timing could be due to the presence of abducens internuclear neurons, which are intercalated between saccade burst neurons and medial rectus motoneurons. The time to traverse the axon of the internuclear neuron, which ascends within the medial longitudinal fasciculus, and its synapse in the oculomotor nucleus might lead to a difference in timing between the activation of the medial and lateral rectus muscles. Simulation of such a delay does produce a transient relative divergence followed by convergence change in intrasaccadic ocular alignment of the correct amplitude, but its temporal profile (Fig. 13, *right*) does not match the experimental data as well as the model that used a difference in abduction-adduction muscle dynamics. We can not exclude some combination of a time

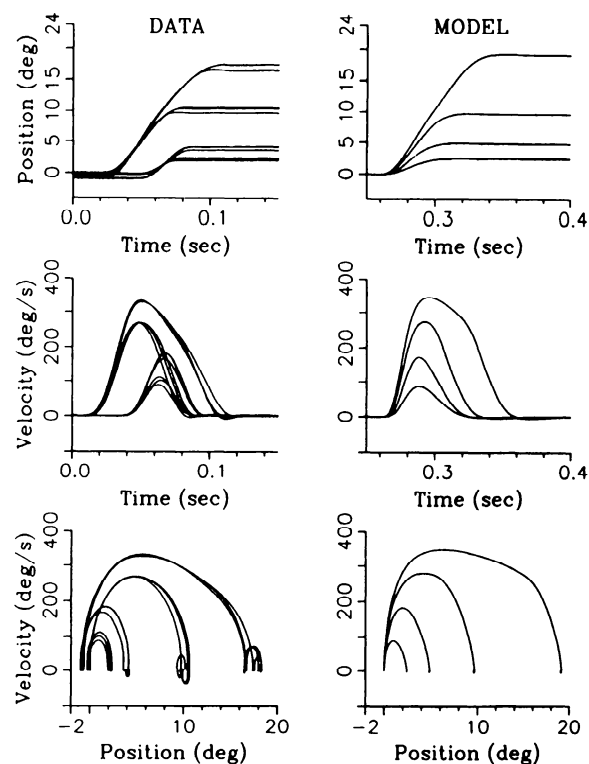


FIG. 12. Comparison of pure saccades generated by the model with the data of *subject 1*. Conjugate position (*top*) and velocity (*middle*) traces are depicted as well as phase plane diagrams (*bottom*). Note how well the simulated and experimental data match. Small reversals in velocity at end of larger saccades in the data of *subject 1* probably represent dynamic overshoots.

¹ All simulations were performed with the use of simulation package (ASP) developed by L. Optican and H. Goldstein. Because the general character of the data sets from the different subjects was similar, we have chosen to simulate in detail the data from *subject 1*. The complete details of the simulation are presented in the APPENDIX, Fig. 19, and Table 1.

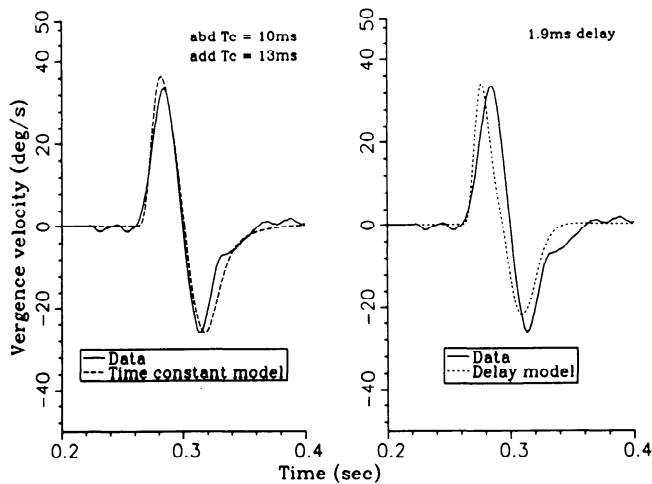


FIG. 13. Simulation of saccade model shown in Fig. 10A and experimental data for the inherent change in horizontal alignment that occurs during pure horizontal saccades for *subject 1*. *Left*: simulation using an abduction-adduction asymmetry in the mechanical properties of the horizontal rectus muscles as reflected in the time constant of the 2nd real pole in the ocular motor plant (time constant of 10 ms for abduction and 13 ms for adduction; see also APPENDIX). *Right*: simulation assuming a small delay (1.9 ms) in signals reaching the medial vs. the lateral rectus muscles.

delay and a difference in muscle mechanics, nor that there are other, more central, factors contributing to the abduction-adduction difference in normal saccades. Nevertheless, the proposed asymmetry in mechanical properties of the horizontal recti is the simplest change that simulates the experimental results satisfactorily.

GENERATION OF PURE VERGENCE MOVEMENTS. We next developed a conceptual scheme for generating pure vergence movements in response to step changes in target depth. We were primarily interested in the motor aspects of vergence movements and so did not consider possible differences in responses to the various types of stimuli that can drive vergence movements, such as disparity, accommodation, and proximity. Our targets actually provided all three types of cues simultaneously.

Local feedback model for vergence. We adopted a suggestion of Zee and Levi (1989), on the basis of the experimental findings of Mays et al. (1986), that the premotor commands for vergence to *step changes* in the position of the target in depth derive from a neural network with a similar structure to that that had been previously developed for the generation of saccades. This idea also evolved from the results of Semmlow et al. (1986), who found that the vergence response to relatively high-velocity ramp disparities can be divided into two components, transient and sustained, that are in many ways analogous to the saccade (open-loop) and pursuit (closed-loop) behavior of conjugate eye movements. Erkelens et al. (1989c) also suggested that disjunctive movements have both saccadic and pursuit components.

For our model we propose that a signal encoding a desired change in vergence angle (Desired ΔV) is compared with an instantaneous efference copy of the actual change in vergence angle ($\Delta V'$), to generate a vergence motor error signal (VME; Fig. 10B). The VME, in turn, drives a set of vergence premotor neurons (vergence velocity neurons or

VVN) to produce a vergence velocity command (VVC). Mays and colleagues (1986) have identified neurons within the midbrain of the rhesus monkey that discharge in relationship to vergence velocity during pure vergence movements. These could be the neural substrate of the hypothesized VVN. We also suggest that the activity of VVN is gated by a set of vergence-related pause neurons (VPN), with the use of a trigger logic analogous to that used for saccades (see also APPENDIX).

The output of the VVN, after mild low-pass filtering (Fig. 10B, filt), is passed directly to the motoneurons as a VVC (comparable with the pulse of innervation for saccades), and indirectly, via a vergence neural integrator (VNI), as a vergence position command, to hold the eyes at the specified vergence angle (comparable with the step of innervation for saccades). The VVC is also passed to the motoneurons through the same filter that produces the slide of innervation for saccades. Studies of the discharge characteristics of medial rectus motoneurons during vergence movements are compatible with this scheme (Gamlin and Mays 1992). The VVC is also fed back and integrated within a vergence local feedback loop (VRI), to provide the efference copy of the change in vergence angle ($\Delta V'$) necessary for the computation of VME. Just as in the local feedback model for saccades, the nonlinear relationship between VME and the output of the VVN (Fig. 11, *VVN panel*), determines the trajectories and the main sequence for vergence eye movements.

To simulate the differences between divergence and convergence, we adjusted the shape of the nonlinearity relating VME and the output of the VVN (Fig. 11, *VVN panel*). To reproduce the exact shape of the vergence waveform, we had to assume that 1) at the onset of discharge of VVN, there is a slight lag (which we attribute to a "recruitment" phenomenon) in the time for VVN to build up to the correct level of activity specified by the VME; and 2) the output of the VVN is low-pass filtered with different time constants for convergence (0.01 s) and divergence (0.05 s).

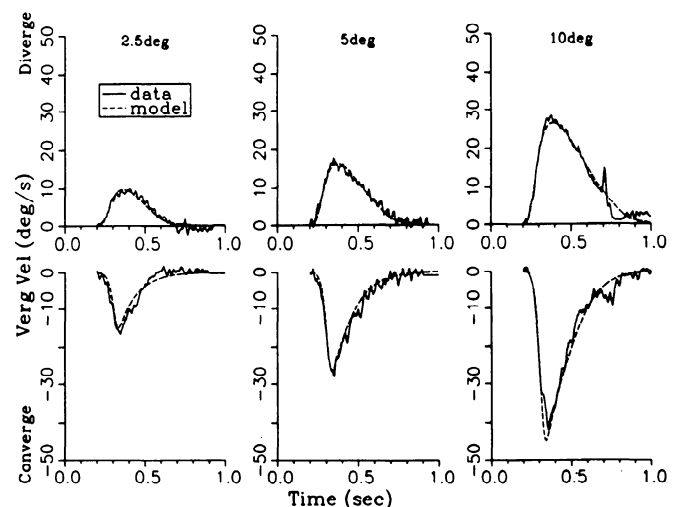


FIG. 14. Simulation of model depicted in Fig. 10B compared with data of *subject 1* for pure vergence responses. Note the close correspondence of the simulated and experimental vergence velocity traces, both with respect to peak values and waveforms. Details of the simulation are in APPENDIX, Fig. 19, and Table 1.

Simulations are compared with normal data from *subject 1* in Fig. 14. By examining vergence velocity, rather than vergence position traces, the quality of the simulations could be more critically evaluated. Just as is the case for saccades and the associated local feedback model, our vergence model, with one particular set of parameters, automatically reproduces the equivalent of the main sequence for pure convergence or divergence movements and also mimics the waveforms of vergence velocity.

GENERATION OF VERGENCE COMBINED WITH SACCADDES. With a model in hand that faithfully simulated both pure saccades, and pure vergence movements to step changes in target depth, we next tackled the more difficult question. When vergence is combined with saccades, why does ocular alignment change faster than would be expected from a simple addition of the change in alignment that occurs during pure saccades and the change that would be produced by a pure vergence movement alone?

Ono et al. (1978) and Kenyon et al. (1980b) suggested that the unexpectedly large difference in the sizes of the movements of the two eyes that occurs when horizontal saccades are combined with vergence results from a nonlinear interaction in the ocular motor plant. We thought this

unlikely to be the major explanation because vertical saccades (which use different muscles) also facilitate horizontal vergence. Consequently, we looked for an explanation for unequal saccades during vergence on the basis of an interaction within the central neural networks that generate premotor saccade and vergence commands. We then developed and simulated several types of hypothetical neural models that could produce such a saccade-vergence interaction.

Omnidirectional pause neurons as the link between saccades and vergence. A critical observation that directed our ideas about an interaction between saccades and vergence in central structures was the finding that horizontal vergence is facilitated during vertical as well as during horizontal saccades. Because the activity of OPN is thought to gate activity of both horizontal and vertical saccadic burst neurons, we wondered if OPN might not also gate activity in neurons generating horizontal vergence commands. Then horizontal vergence could be facilitated during saccades of any direction. With this idea in mind, we developed several different types of models of saccade-vergence interaction, although in each type OPN play the pivotal role in the facilitation of vergence by saccades.

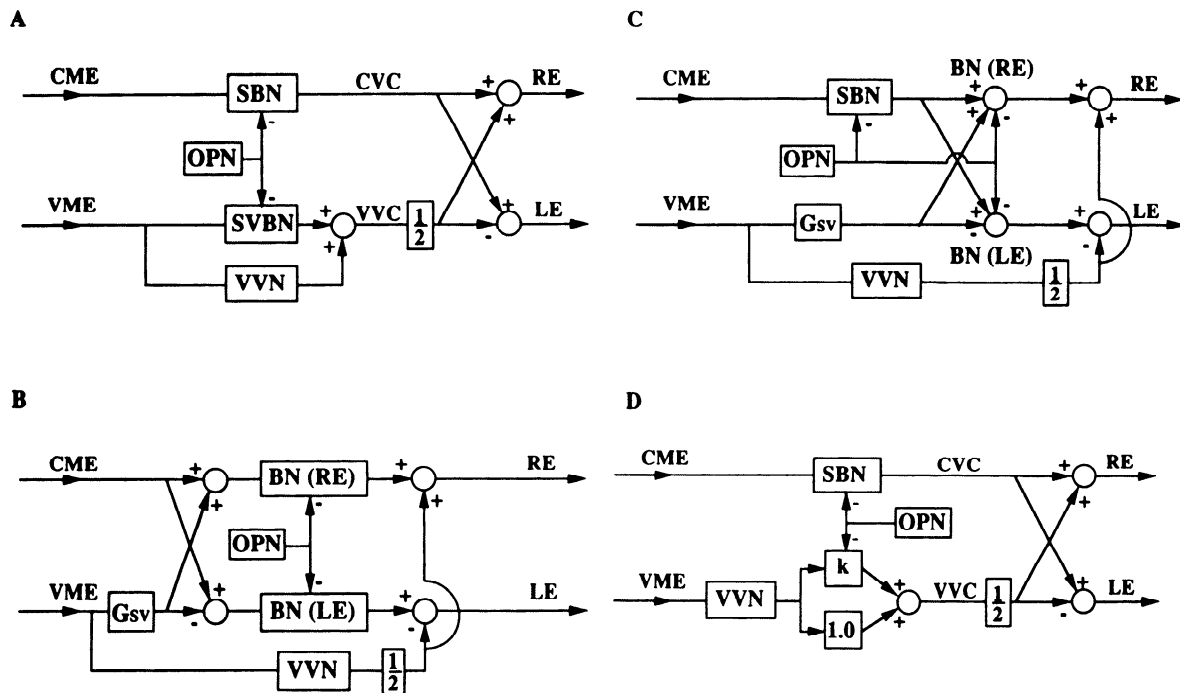


FIG. 15. *A:* SVBN (saccade-related vergence burst neuron) model. Separate saccade and vergence-related burst neurons. Omnidirectional pause neurons (OPN) gate the activity of both saccade burst neurons (SBN) and saccade-related vergence burst neurons (SVBN). The outputs of the saccade-related vergence burst neurons (SVBN) and the vergence velocity neurons (VVN) are added to produce a vergence velocity command (VVC). CVC, conjugate velocity command; RE and LE refer to velocity commands; CME, conjugate motor error; VME, vergence motor error. *B:* DB (difference burst) model. VME signals (of opposite signs) and CME signals (of the same sign) are summed on separate right eye and left eye burst neurons [BN(RE) and BN(LE)]. Before summation the VME signals are multiplied by the appropriate gain (Gsv) to produce differences between divergence and convergence. Pure vergence commands from vergence velocity neurons (VVN) are summed with the outputs of the right eye and left eye pulse generators. *C:* modified DB model. A modified version of the DB model in which a cyclopean saccade burst is 1st created on conjugate burst neurons (SBN) and then passed to a 2nd set of burst neurons [BN(RE) and BN(LE)] that project separately to either the left eye (LE) or the right eye (RE). VME signals are multiplied by the appropriate gain (Gsv) to produce differences between divergence and convergence, and then added (with opposite signs) to the separate right eye and left eye burst neurons. Both sets of burst neurons are under pause cell (OPN) inhibitory control. *D:* Multiply model. In this model OPN partially inhibit the activity of VVN so that during a saccade, when OPN inhibition is completely removed, the gain of VVN increases from 1.0 to $K + 1.0$. Not shown in these diagrams are the vergence pause neurons (VPN), which gate activity within VVN (see Figs. 10B and 19).

SACCADE-RELATED VERGENCE BURST NEURON (SVBN) MODEL. In our first model (Fig. 15*A*, SVBN), we suggest that the activity of OPN gate the activity of a hypothetical, separate class of neurons, to be called SVBN. SVBN generate pre-motor horizontal vergence commands, but only during a coincident horizontal or vertical saccade. The output of this saccade-related vergence pathway (SVBN) sums with the output of the pure vergence pathway (VFN) and so acts to increase vergence velocity selectively during saccades. The SVBN are posited to be driven by the same VME that drives the VFN, but the shape of the relationship between the VME and the activity of the SVBN and of the VFN differ (Fig. 11, compare *middle* and *right panels*). Note also that the nonlinearities have been adjusted separately for divergence and convergence. The output of the SVBN is summed with the output of the VFN to produce the composite VVC. Again, the VVC and its filtered and integrated versions are passed to the final common pathway, and the

VVC is also passed back and integrated in the vergence local feedback loop for the calculation of the efference copy of vergence angle.

By empirically adjusting the shape of the SVBN nonlinearity (Fig. 11, *right panel*), we found one set of parameters for divergence and one for convergence that enabled us to simulate successfully almost all combinations of saccades and vergence with 2.5, 5, and 10° stimulus amplitudes (the range of values measured experimentally). Examples for 10° saccades with 5° vergence, and 5° saccades with 2.5° vergence are shown in Fig. 16, *A* and *B*, respectively. Note how closely the simulated and experimental waveforms of vergence velocity match each other. The model simulation even produces the subtle inflections in the vergence velocity waveforms that appear near the beginning and the end of the saccade. Simulations of the entire data set for *subject 1* are shown in the APPENDIX (Figs. 20 and 21). For comparison, Fig. 18, *far right*, shows the simulation of a model in which there is no interaction, i.e., a simple addition of the pure vergence response and the pure saccade response. This “no interaction” simulation does not account for the actual change in alignment that occurs during vergence combined with saccades (also compare with experimental data, Fig. 6, *right panels*).

Generation of horizontal vergence with vertical saccades. We next addressed the issue of facilitation of vergence by vertical saccades. Again, it is necessary to take into account the transient changes in horizontal alignment that occur during pure vertical saccades. In general, pure upward saccades were associated with an initial divergence and downward saccades with an initial convergence, as reported before (Collewijn et al. 1988b; Enright 1989). Unlike for horizontal saccades, however, we have no ready hypothesis to account for the changes in horizontal alignment that occur during vertical saccades. Thus changes in horizontal alignment during pure vertical saccades were not simulated, but rather the actual change in horizontal alignment that occurred during pure vertical saccades made by the subject was added to the model output. We justified this approach because our major goal was to explore the mechanism by which vertical saccades facilitate horizontal vergence, not the mechanism for the transient changes in horizontal alignment that occur during pure vertical saccades themselves.

To simulate the vergence response to a horizontal disparity combined with a vertical saccade, the omnidirectional pause cells were turned off for the equivalent duration of a vertical saccade so that the SVBN could be activated during the time that the vertical saccade would have been occurring. Examples of simulated and experimental data are shown in Fig. 17*A*. The match is quite good.

Some degree of trial-by-trial variability in the facilitation of horizontal vergence by vertical saccades was a feature of our data. We could simulate this finding by varying the degree to which SVBN became engaged during vertical saccades. We suggest that the release of SVBN from inhibition by OPN is not complete for every vertical saccade; this variability would account for the trial-to-trial fluctuation in the amount of facilitation of vergence.

We also found that conjugate horizontal saccadic oscillations, which often appeared during pure vertical saccades as

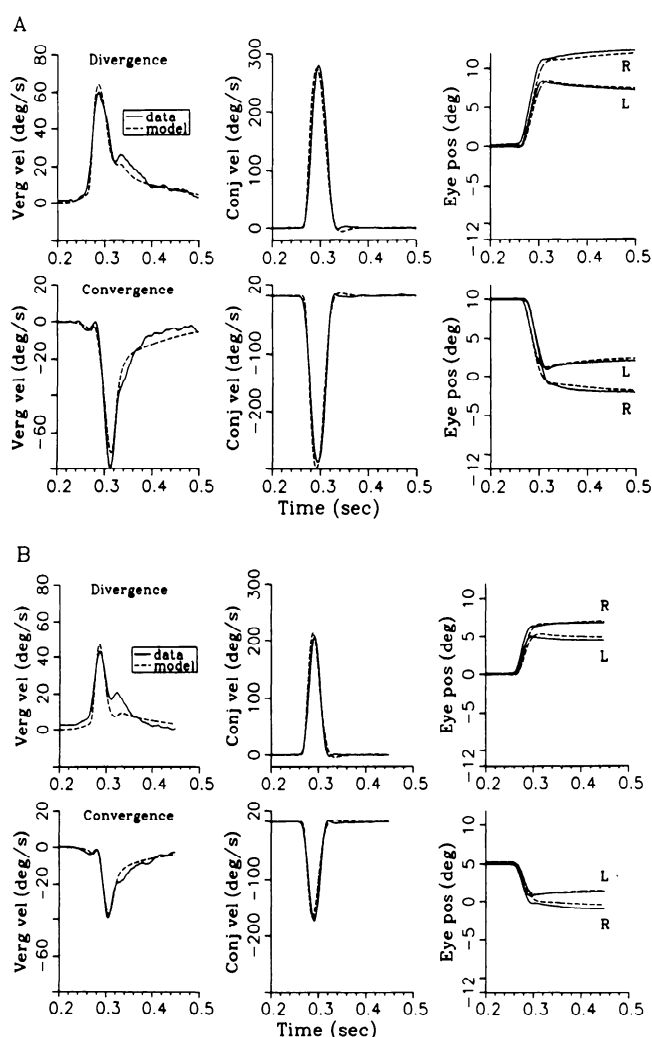


FIG. 16. Comparison of simulations of the saccade-related vergence burst neuron (SVBN) model with data of *subject 1* for 10° horizontal saccades combined with 5° vergence (*A*) and for 5° horizontal saccades combined with 2.5° vergence (*B*). From *left to right* the panels depict vergence velocity, horizontal conjugate velocity, and the horizontal positions of the left and right eyes. Simulations of both position and velocity match the experimental data quite well, and in particular note that the simulations reproduce most of the subtle nuances in the vergence velocity waveforms.

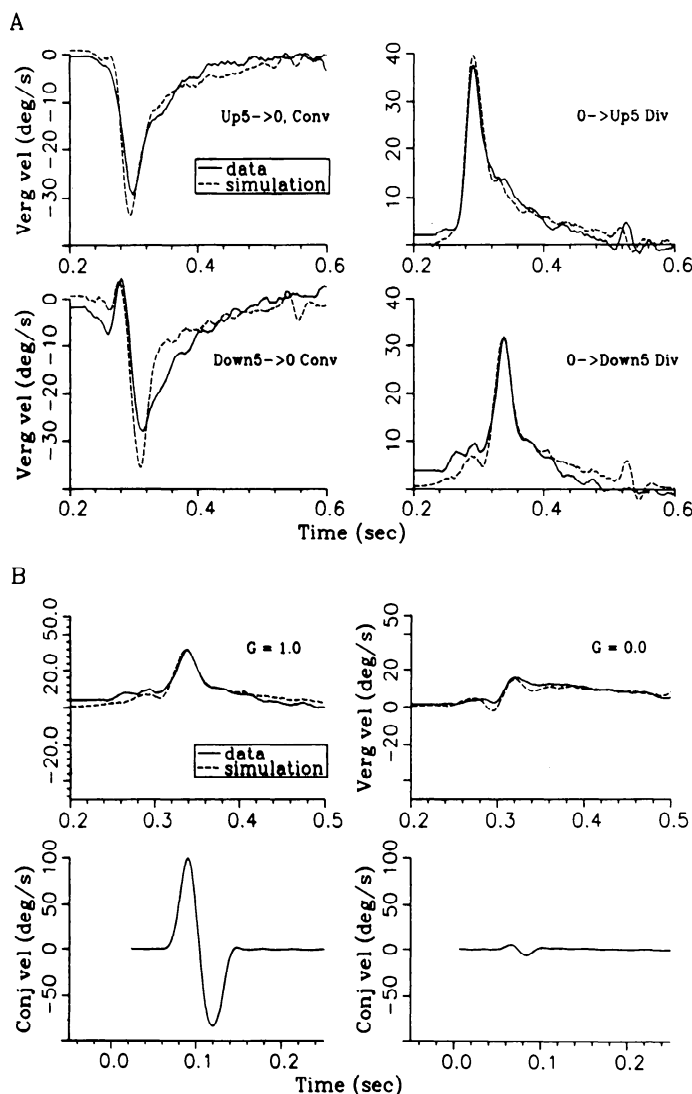


FIG. 17. *A*: comparison of simulations of the saccade-related vergence burst neuron (SVBN) model with data of *subject 1* during 5° vertical saccades with 2.5° of vergence. Responses with saccades made between 0 and Up 5° , and between 0 and Down 5° are shown. Note that the inherent change in horizontal alignment that occurred during pure vertical saccades made by *subject 1* was added to the output of the model to produce the simulation. *B*: comparison of simulations of the SVBN model with 2 trials from *subject 1* for 5° vertical saccades (from 0 to Down 5°) combined with 2.5° of divergence. For the trial depicted in *left panels*, the horizontal vergence velocity during the saccade and the conjugate horizontal oscillations are much larger than for the trial shown in *right panels*. *Left-hand data* could be simulated with the SVBN model ($G = 1.0$), whereas *right-hand data* was simulated assuming only a linear summation of a pure vergence and pure saccade command, i.e., no interaction ($G = 0.0$). Note that the inherent change in horizontal alignment that occurred during pure vertical saccades made by *subject 1* was added to the output of the model to produce the simulations.

well as during vertical saccades with horizontal vergence, were often larger when horizontal vergence was more strongly facilitated (Fig. 17*B*). Our explanation for the relationship between conjugate saccadic oscillations and vergence facilitation is as follows. Because of presumed delays in the local feedback loop around the saccade burst neurons, and the inherently high gain of the motor error-saccade burst neuron relationship, the saccade pulse generator

is inherently unstable and susceptible to oscillations (Ashe et al. 1991; Zee and Robinson 1979). Thus one might expect conjugate horizontal oscillations during pure vertical saccades and vice versa, because the pause cells decrease their discharge for saccades made in any direction. Conjugate oscillations (or a half cycle of oscillation) on the horizontal trace were often observed during pure vertical saccades and were of varying amplitude. On the basis of these observations, we suggest some variability in the degree to which horizontal saccade burst neurons (and saccade-related vergence burst neurons) are completely disinhibited by OPN during pure vertical saccades. Likewise, during horizontal vergence combined with vertical saccades, one would be more likely to observe both large-amplitude conjugate horizontal saccadic oscillations and better facilitation of horizontal vergence, when all of the OPN completely cease discharging.

To epitomize, the SVBN model has the following advantages. By virtue of the OPN link, the model explains the facilitation of horizontal vergence by both horizontal and vertical saccades. With relatively few assumptions, it automatically accounts for the main sequence relationships of both pure vergence eye movements and a variety of combinations of vergence eye movements and saccades of different sizes. At the heart of the SVBN model are the nonlinear functions describing the relationships between VME and the output of the pure VVN and of the SVBN. Correct choices for these nonlinearities allow the model to simulate not only the gross behavior of pure vergence and of vergence combined with saccades, but also the subtle characteristics of the vergence velocity waveforms for a wide variety of stimulus combinations. This degree of parsimony makes the SVBN model attractive in the same sense as does the local feedback model for the generation of saccades (Robinson 1975; Zee et al. 1976).

DIFFERENCE BURST (DB) MODEL WITH SEPARATE RIGHT EYE AND LEFT EYE SACCADIC PULSE GENERATORS. In our second model (Fig. 15*B*, DB) we suggest that there are *separate* groups of saccade burst neurons for the right and left eyes, in addition to the well-accepted notion of separate groups of saccade burst neurons for the right and left and the up and down *directions*. If there were separate right and left eye burst neurons, conjugate or cyclopean motor error signals could be added with the *same* sign, and vergence motor error signals with *opposite* signs, onto the separate right and left eye burst neurons. The somewhat heretical suggestion of separate pulse generators for each eye implies that saccades made by each eye could be under independent control. There is some, albeit scanty, evidence for this proposition. Normal subjects occasionally make disjunctive (oppositely directed) saccades during attempted pure vergence (Levi et al. 1987), and disjunctive saccades have been described in rabbits and birds (Bloch et al. 1987; Collewijn and Noorduyn 1972; Wallman and Pettigrew 1985). Similarly, patients with neurological disease may show a nystagmus with disjunctive saccades (Ochs et al. 1979; Yee et al. 1979). Although rare, the fact that disjunctive saccades occur at all implies the existence of an underlying physiological substrate for separate control of saccades made by each eye. Thus it is useful to explore the idea of separate right

and left eye saccade generators and their possible interaction with vergence.

If there were separate right and left eye burst neurons, then CME signals could be added with the *same* sign, and VME signals with *opposite* signs, onto the separate right and left eye burst neurons (Fig. 15B). In this scheme, OPN would still be able to gate vergence commands selectively during saccades, and so facilitate vergence during saccades. The saccade-related component of the VVC would now be reflected in the difference burst between the outputs of the right and left eye burst neurons. The amplitude of the difference burst could be adjusted by multiplying the VME by the appropriate gains for divergence and convergence without altering the waveform of the saccades.

On scrutiny of the DB model, however, one important difference from the SVBN model immediately emerges. The amplitude of the difference burst for a given VME depends not only on the amplitude of the VME itself but also on the amplitude of the conjugate motor error. This interaction occurs because the values from which the difference burst is calculated are derived from the nonlinear relationship relating the motor error to saccade burst neuron discharge for the left and right eye saccade pulse generators. Hence the shape of the motor error; burst output nonlinearity for saccades becomes important in determining the value of the difference burst and, consequently, the saccade-related boost in vergence speed.

Because the shape of the left eye and right eye burst neuron nonlinearities determines the dynamic properties of the saccades, the nonlinearities can not be modified to allow for a better simulation of vergence during saccades, without altering the saccades themselves. In other words, the DB model is constrained by the relationship between conjugate motor error and saccade burst neuron discharge. For example, the difference burst must necessarily be small at the beginning of the saccade, when the slope of the nonlinearity for the burst neurons is relatively shallow and must necessarily become larger as the saccade nears completion, when the slope of the nonlinearity becomes relatively steep. This pattern causes the vergence velocity waveform to depart from that of the experimental data (Fig. 18, compare data and DB panels). The failure of the DB model reflects its lack of flexibility; one must be able to shape the output of the saccade-related vergence pathway so that it can interact correctly with the inherent changes in ocular alignment that can be attributed to the saccades themselves.

One possible way to salvage the hypothesis of separate right eye and left eye burst neurons is to assume that the VME is added and subtracted on individual right eye and left eye burst neurons, *after* separate conjugate pulses for each eye have already been created by cyclopean burst neurons (Fig. 15C, modified DB model). The activity of both classes of burst neurons (individual eye and cyclopean) would be under the inhibitory control of and hence gated by OPN. The separate right eye and left eye burst neurons, however, would show a linear relationship between their input and output. In this way, the difference burst no longer needs to be related to the size of the conjugate motor error, yet the dynamic properties of conjugate saccades are unaffected by the second set of individual eye burst neurons (Fig. 15C). The difference burst can also be adjusted with-

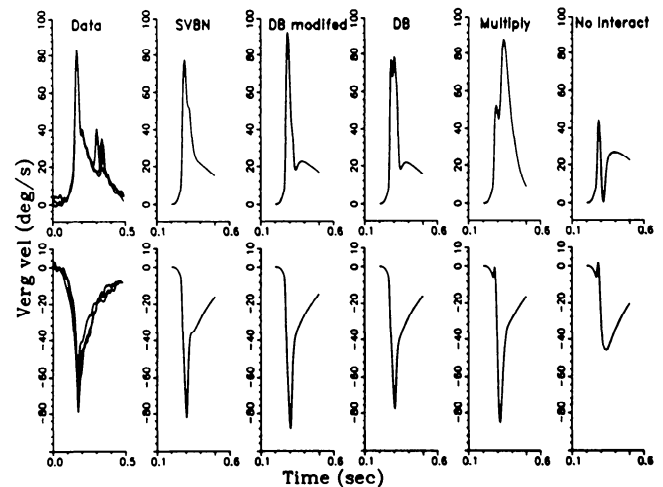


FIG. 18. Comparison of vergence velocity during saccades combined with 10° of vergence for the experimental data of *subject 1* (far left panel, *S1*), with the simulations of the saccade-related vergence burst neuron (SVBN) model, the modified difference burst (DB) model, the DB model, the Multiply model and the no-interaction model. For each model, the simulation was optimized to produce the vergence-saccade interaction that best approximated the experimental data. Overall, the experimental data were best simulated by the SVBN model. The size of the saccade with divergence was 8° ; with convergence, 3° .

out affecting saccades by multiplying the VME by appropriate gains for divergence and convergence. Even with the addition of a second set of burst neurons, however, the DB model is still lacking in flexibility. The saccade-related VVC can not be shaped with enough exactness to interact correctly with the inherent change in alignment that arises from the saccade itself. Although better than the original DB model, the modified DB model still does not match the experimental data as closely as does the SVBN model (Fig. 18, compare SVBN and modified DB panels).

OTHER POSSIBLE NONLINEAR INTERACTIONS. Finally, one must ask if facilitation of vergence during saccades is related to a relatively nonspecific nonlinear interaction, either in central premotor structures, or in the ocular motor plant, at the motoneurons or within the eye muscles themselves. Kenyon and Stark (1983) elaborated on this hypothesis and claimed that their nonlinear model of the ocular motor plant accounted for the large difference between the amplitudes of the movements of the two eyes when saccades and vergence were combined. Kenyon and Stark, however, did not directly address or simulate two crucial observations: the transient changes in ocular alignment that occur during pure horizontal saccades, and the facilitation of horizontal vergence by vertical saccades. Furthermore, the existence of a nonspecific peripheral nonlinear interaction between vergence and conjugate commands is contradicted by the finding that during vergence combined with pursuit there seems to be relatively little deviation from linearity; only an 11% *slowing* was noted, and then only in the eye in which the vergence and pursuit movements were in the same direction (Miller et al. 1980).

Nevertheless, we tested the idea of a nonspecific nonlinear interaction in central structures as a cause of saccade-related facilitation of vergence, by increasing the gain of the pure vergence pathway (VVN) selectively during saccades

(Fig. 15D, Multiply model). One possible mechanism for this type of interaction has recently been suggested by Mays et al. (1992), who found that stimulation of the OPN region during pure vergence movements resulted in a decrease in vergence velocity. If during fixation, OPN partially inhibited not only the saccade burst neurons but also the pure VVN, then, when a saccade did occur, the inhibition of VVN by OPN would be lifted, and the VVN could discharge at a higher frequency for a given VME signal. This increase in the gain of VVN during saccades would lead to a saccade-related increase in vergence velocity. Simulation of this simple model did not match our experimental data as well as the SVBN model (Fig. 18, compare Multiply with SVBN model).

One could improve, however, the Multiply model by shaping the multiplicative nonlinearity itself, in a way that was similar to the shaping of the nonlinearity for the posited saccade-related vergence burst neurons in the SVBN model. Adopting this strategy would make the Multiply model indistinguishable from the SVBN model and, in effect, would turn VVN into SVBN during saccades. There is not as yet, however, enough physiological evidence to distinguish clearly between the SVBN and the Multiply models.

Model predictions

SINGLE-UNIT BEHAVIOR IN EXPERIMENTAL ANIMALS. Our computer simulations of the interaction between saccades and vergence lead to a number of predictions, both as to patterns of neural firing of vergence-related neurons within the brain stem, and to certain aspects of the behavior of vergence eye movements. First, the inherent change in alignment that occurs during pure saccades, which we suggest arises from peripheral mechanical asymmetries in the ocular motor plant, should not be reflected in the activity of premotor vergence neurons. Secondly, for the SVBN model (Fig. 15A), there should be distinct saccade-related vergence velocity neurons (SVBN) and pure VVN. SVBN should discharge in association with horizontal or vertical saccades, but only when there is a coincident vergence movement. For saccades of a given amplitude, the peak discharge of SVBN should increase with the amplitude of vergence. For vergence of a given amplitude, the peak discharge of SVBN should be nearly constant. SVBN should not discharge during pure saccades nor during pure vergence movements.

Pure VVN should discharge with all vergence eye movements, whether or not they are associated with saccades. During pure vergence movements, VVN should show an increase in peak discharge as the amplitude of vergence increases. During saccades combined with vergence, the pattern of discharge of VVN should differ slightly from that during pure vergence movements of the same amplitude. This is because the input to the VVN, which is VME, is determined by the sum of the output from VVN and SVBN. VVN should not discharge during pure saccades. The outputs of the two types of vergence velocity neurons, VVN and SVBN, could be combined on a set of more immediately premotor vergence neurons that would discharge in proportion to vergence velocity whether or not a saccade was combined with the vergence movement. In contrast, if

the Multiply model is correct (Fig. 15D), there should only be one class of premotor VVN, and their activity should always be related to vergence velocity.

On the other hand, if there are separate right eye and left eye saccade burst neurons that also receive VME signals (DB model, Fig. 15B), one should be able to find individual burst neurons that discharge more closely with the movements of one eye than with the other, when saccades are combined with vergence. If the modified version of the DB model is correct, in which the VME is added to right eye and left eye burst neurons after the conjugate saccade command is created (Fig. 15C), one should be able to distinguish two different classes of saccade burst neurons. The discharge rate of one type should always be tied to the conjugate, cyclopean saccade, even when the saccade is combined with vergence, and the discharge rate of the other type should always be tied to the movements of either one or the other eyes when saccades are combined with vergence. Fur-

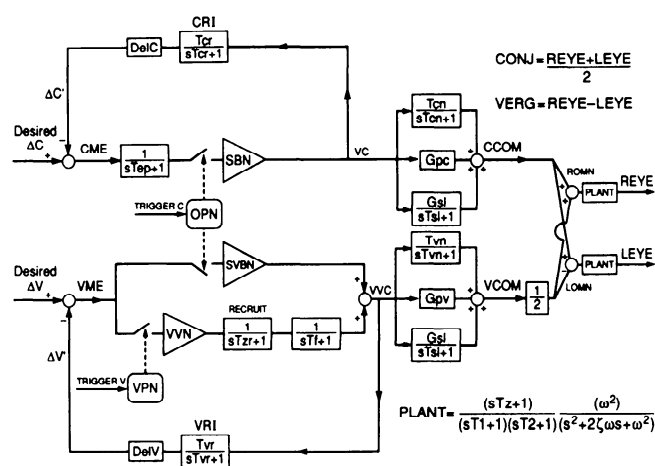


FIG. 19. Block diagram of the SVBN model. Specific values and additional definitions of parameters are specified in Table 1. Time between iterations of the simulation was 0.0002 s. Equations describing the nonlinear relationships between conjugate motor error (CME) and the output of the saccade burst neurons (SBN), and between vergence motor error (VME) and the output of vergence velocity neurons (VVN) and of saccade-related vergence burst neurons (SVBN), are in Table 1. Triangular (amplifier) symbols signify the types of neurons on which the nonlinearities are created. Specific values of the parameters for the nonlinearities are shown in Table 1, and the curves for the nonlinearities are plotted in Fig. 11. Recruit, recruitment lag for VVN. TriggerC and OPN (omnidirectional pause neurons) refer to the gating logic that allowed SBN and SVBN to discharge during saccades and prevented them from discharging when a saccade was not being made. Similar logic was used to control the action of vergence velocity neurons (VVN) by vergence pause neurons (VPN and TriggerV). Not shown on the diagram is a small lag (time constant of 0.001 s) to smooth the switching of the 2nd pole time constant between abduction (0.010 s) and adduction (0.013 s) when the eye changes horizontal direction. Also not explicitly shown is that the input command signals (Desired ΔV and Desired ΔC) to premotor vergence and saccade neurons were derived from an estimate of the position of the target (both in depth and across the visual field) with respect to the head. Calculation was based on the difference between and on the average of the retinal error signals for each eye, which were then added to efference copy estimates of the conjugate and vergence eye positions. VCOM, vergence command. CCOM, conjugate command; ROMN and LOMN, right and left eye ocular motor neurons. REYE and LEYE, right and left eye positions; CONJ, conjugate eye position; VERG, vergence eye position; CRI, conjugate resettable integrator; VRI, vergence resettable integrator; s, Laplace transform operator. Other parameters and abbreviations are defined in Table 1.

thermore, anatomic studies should show separate projections of the hypothesized right eye and left eye burst neurons to separate pools of neurons (abducens internuclear neurons and abducens motoneurons) within the abducens nucleus.

TABLE 1. *SVBN Model*

	Parameter	Value
<i>Saccadic system</i>		
Gain of conjugate pulse	Gpc	0.06
Gain of slide	Gsl	0.165
Tc slide	Tsl	0.08 s
Tc conjugate neural integrator	Tcn	20 s
Tc motor error pole	Tep	0.003 s
Tc conjugate resettable integrator	Tcr	20 s
Saccade local feedback delay	DelC	0.006 s
SBN nonlinearity		
$yR = AR \{1.0 - \exp[-(e0 + x)/\lambda R]\}$	e0	5
$yL = AL \{1.0 - \exp[-(e0 - x)/\lambda L]\}$	AL	440
for $-e0 < x < e0$	AR	400
$y = yR - yL$	λL	10
	λR	10
<i>Plant</i>		
	Tz	0.08 s
	T1	0.3 s
	T2	0.010 s abd
		0.013 s add
	ω	200 radians/s
	ζ	1.2
<i>Vergence system</i>		
Gain of vergence pulse	Gpv (conv)	0.08
	Gpv (div)	0.01
Tc vergence neural integrator	Tvn	10 s
Vergence local feedback delay	DelV	0.003 s
Tc vergence resettable integrator	Tvr	20 s
VVN nonlinearity		
$y = kx^n/(x^n + \theta^n)$	nC	1.0
	nD	0.8
	kC	120
	kD	125
	θC	15
	θD	30
Recruitment lag		
$Tzr = (k1)(k2)$	Tr1	0.02 s
$K1 = VPAUSE/[(sTr1 + 1)(sTr2 + 1)]$	Tr2	0.005 s
	k2 (conv)	1.0
	k2 (div)	0.1
VVN filter	Tf (conv)	0.01 s
	Tf (div)	0.05 s
SVBN nonlinearity		
$y = A[1.0 - \exp(-x/\lambda)]$	AC	60
	AD	80
	λC	6
	λD	4

Values and definitions for the parameters used in the simulation of the SVBN model shown in Fig. 19. The parameters for the equations describing the nonlinearities for the saccade and vergence burst neurons (SBN, VVN, SVBN) are separated into right and left, designated by values with an appended R or L, and into divergence and convergence, designated by values with an appended D or C, or div or conv. The actual equations for the nonlinearities and the recruitment lag are shown in the table. The equation for the ocular motor plant is shown on Fig. 19. SVBN, saccade-related vergence burst neuron; SBN, saccade burst neuron; VVN, vergence velocity neuron. Tc, time constant; VPAUSE, state of vergence pause neurons and switches between 1.0 (no vergence) and 0.0 (vergence).

All of our models predict that artificial stimulation of omnidirectional pause cells should influence the trajectory of ongoing vergence movements. For the SVBN model, vergence associated with saccades should be altered in the case of stimulation of OPN, and vergence with or without saccades, in the case of stimulation of the proposed VPN. For the Multiply model, stimulation of OPN should accomplish both, i.e., altering both pure and saccade-related vergence. Our simulations also predict that there should be some variability in the degree to which OPN decrease their discharge rate during *pure* vertical saccades. This would account for the variable degree to which horizontal vergence is facilitated during vertical saccades. Finally, if OPN do influence vergence eye movements, there should be anatomic projections of OPN to the central mesencephalic reticular formation, in which vergence-related premotor neurons are found. There is evidence that this is the case (Büttner-Ennever and Büttner 1988).

BEHAVIORAL OBSERVATIONS. If the idea that vergence and saccades are linked by pause cell activity is correct, then blinks of the eyelids, too, might be expected to have some influence on vergence movements. Blinks, even without eye movements, are associated with a decrease in pause cell discharge, and may cause saccadic oscillations (Hain et al. 1986). Thus one might expect blinks to facilitate vergence movements, and there is some evidence that this is indeed the case (e.g., Peli and McCormack 1986).

There is also some evidence that accommodative vergence is facilitated by saccades (Enright 1986). Our model could easily reproduce this finding because the input command signal, desired change in vergence angle, is derived from the perceived distance of the target (expressed as a vergence angle, see legend of Fig. 19). The calculation of the perceived distance of a target could use accommodation and proximity as well as disparity cues.

Finally, our models have implications for the process of (disconjugate) adaptation to conditions that require an adjustment in the relative innervation to the two eyes during saccades. Examples include the adaptive response to wearing optical aids such as an anisometropic spectacle correction or afocal magnifiers (Erkelens et al. 1989a; Lemij and Collewyn 1991a,b; Oohira and Zee 1991; Schor et al. 1990; Zee and Levi 1989) or to small degrees of asymmetry in muscle strength (Viirre et al. 1988). In these circumstances the brain learns to program saccades of different sizes, independent of any immediate presence of depth cues. Our hypothesized circuits for producing unequal saccades, and in particular the activity of the SVBN, could be easily modified to link a change in ocular alignment automatically with saccades, even in the absence of disparity cues.

Functional implications

From the point of view of teleology, the close relationship between saccades and vergence is not surprising because they both work toward optimizing the same aspect of visuomotor performance: rapid identification and localization of new objects appearing away from the point of regard. Saccades are required for the early identification of a new target, and vergence for a more precise estimate of the position and the configuration of the target in three dimensions. This close relationship between saccades and ver-

gence is perhaps reflected in the phenomenon of “saccadic suppression”; the relatively small elevation of the threshold for the detection of a flash of light during saccades made in otherwise dark surroundings. A similar elevation of threshold has been described for pure vergence to steps of disparity (Hung et al. 1989).

Finally, we should reemphasize that our model is designed to account for the rapid vergence responses to step

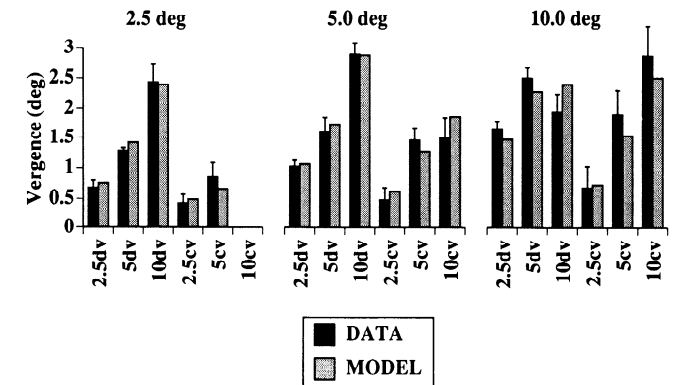
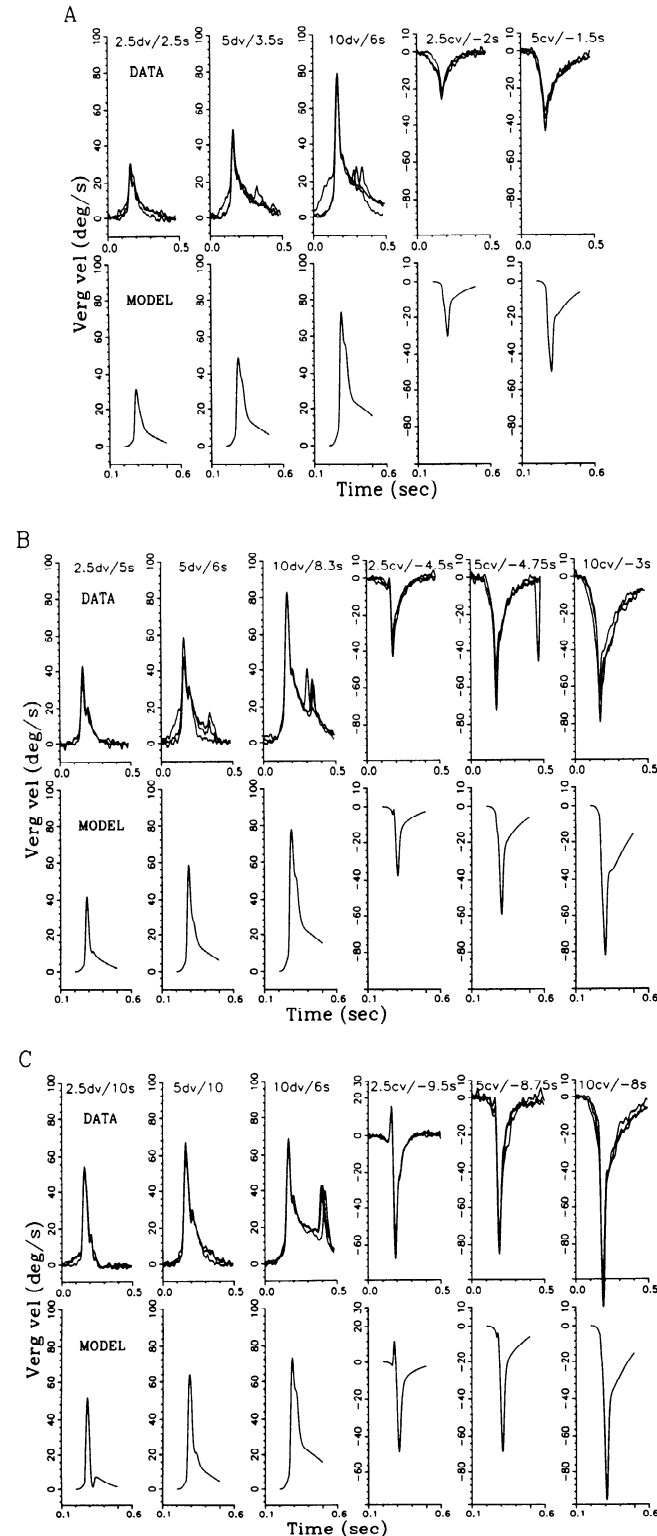


FIG. 21. Histograms of the amplitude of the change in intrasaccadic alignment for data of *subject 1* (■) and for the saccade-related vergence burst neuron (SVBN) model (▨). The data set used here is the same as depicted in Fig. 20. For the experimental data, $n \geq 9$ for all trial types except 2.5 saccade with 5° convergence ($n = 3$) and 2.5 saccade with 10° convergence ($n = 0$). Error bars above experimental data indicate standard deviation for each trial type. They ranged from 4 to 51% of the mean value with an average of 19%. Plotted here are the actual changes in alignment during saccades combined with vergence (i.e., not corrected for any inherent changes in alignment during pure saccades, as was used to calculate the histograms shown in Fig. 9). The end of the saccade generated by the simulations (equivalent to the placement of the “p” on the experimental data) was set when the absolute value of conjugate eye velocity became $<40^\circ/\text{s}$. Note the close correspondence between the experimental and simulated data.

changes in target depth, not for the sustained vergence responses to targets that are moving more slowly and smoothly in depth. It is not unreasonable, however, to speculate that a mechanism analogous to that driving conjugate smooth pursuit of targets moving slowly across the visual field might generate the slower, sustained vergence response to targets that are moving smoothly in depth.

APPENDIX

The detailed diagram of the SVBN model is depicted in Fig. 19, and its parameters are defined and specified in the legend of Fig. 19 and in Table 1. Figure 20 shows the close match between the simulations of the vergence velocity waveforms and those of the experimental data, for a variety of combinations of vergence and saccades. For consistency, all velocity signals from the simulation were obtained by applying a finite impulse response (FIR) filter in the same way as for the experimental data. Figure 21 shows the quantitative agreement between model and experimental data in the amount of change in alignment during the saccade for the same data set and simulations shown in Fig. 20. The close correspondence of the simulated and the experi-

FIG. 20. Simulation of the entire data set for combined saccade-vergence movements for *subject 1*. Vergence velocity is shown for the combined movements. *Top traces*: experimental data. *Bottom traces*: simulations. *A*: 2.5° saccades with vergence. *B*: 5.0° saccades with vergence. *C*: 10.0° saccades with vergence. No data were available for 2.5° saccades with 10° convergence. Numbers above each *top trace* show the value of the vergence demand (which was always quite close to the actual vergence produced; dv, divergence; cv, convergence) and the mean value of the size of the saccade that was actually made by the subject. Simulations were adjusted according to these values. Note that peak amplitudes and shapes of the waveforms of vergence velocity of the simulated and experimental data match each other closely. Small spikes late in the vergence velocity traces reflect secondary corrective saccades.

mental pure vergence responses was already shown in Fig. 14.

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